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DiffusionTunes
Long-Form Music Generation via Dual Conditioning

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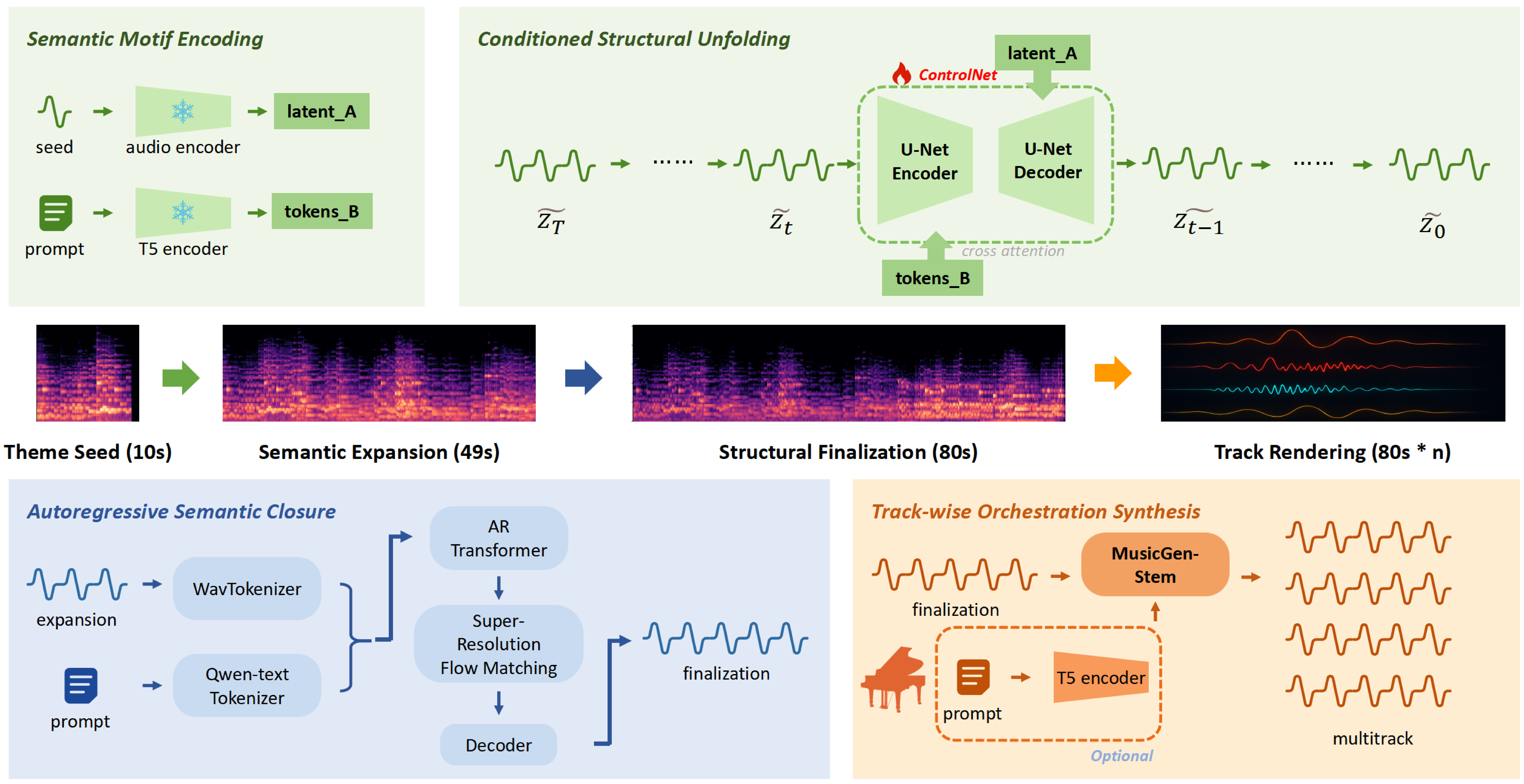

1 Introduction

Recent advances in generative AI have opened new possibilities in music creation, yet generating long-form, structured, and controllable compositions from minimal input remains challenging. Most existing models focus on short clips and lack fine-grained control over musical structure, melody, and instrumentation.

We propose **DiffusionTunes** that aims to bridge this gap by developing a three-stage framework that transforms a short melodic seed into a complete multi-track composition. By integrating diffusion-based generation, autoregressive semantic modeling, and orchestration synthesis, our system enables users to control musical style, structure, and instrumentation through both melody and natural language promptspushing toward expressive, user-controllable, and high-fidelity music generation.

Figure 1: Logo of DiffusionTunes.

2 Methodology

We present a three-stage architecture for controllable music generation that transforms melodic seeds provided by users into complete multi-track compositions. Our system progressively expands short musical motifs through hierarchical semantic and structural modeling, enabling precise control over pattern and style throughout the generation process.

Stage 1: Diffusion-based Main Content Generation

Our approach begins by encoding the input seed melody through a pretrained audio encoder and the prompt through a T5 encoder, producing melodic latent representation and contextual tokens. The melodic latent is then processed through a ControlNet guided denoising framework, where the ControlNet is specifically trained on the Maestro dataset to ensure musically coherent generation, while contextual tokens provides semantic guidance through cross-attention mechanisms. This stage progressively unfolds the musical structure over multiple denoising steps, extending the 10-second input into a coherent 49-second expansion while maintaining harmony and consistency throughout the generation process.

Stage 2: Autoregressive Semantic Closure

Since diffusion-based music generation has limited flexibility in output length control and often produces incomplete endings, we implement an autoregressive framework combining Transformer architecture with super-resolution flow matching to achieve longer-form musical coherence and address these structural limitations, following InspireMusic [3]. This stage extends the composition to 80 seconds while preserving both local musical patterns and global structural integrity, providing complete and satisfying musical conclusions.

Stage 3: Track-wise Orchestration Synthesis

The final orchestration stage utilizes MusicGen-Stem [2] conditioned on the original text prompt to synthesize individual instrumental tracks from the finalized musical structure. This approach enables multi-track composition where each instrument can be generated independently while maintaining harmonic consistency across the ensemble.

3 Results

We evaluate **DiffusionTunes** on three categories of metrices:

- Harmony/Pitch, Phythm, Timbre/Sound, Dynamics, Novelty: Musical quality and structure.
- Continuation: Similarity between the input seed and generated result.
- Length: Length of the generated audio.

We measure the performance of **DiffusionTunes** at three representative steps, it is shown that a tradeoff exists between the quality of generated music and the guidance continuation, as shown in Table 1.

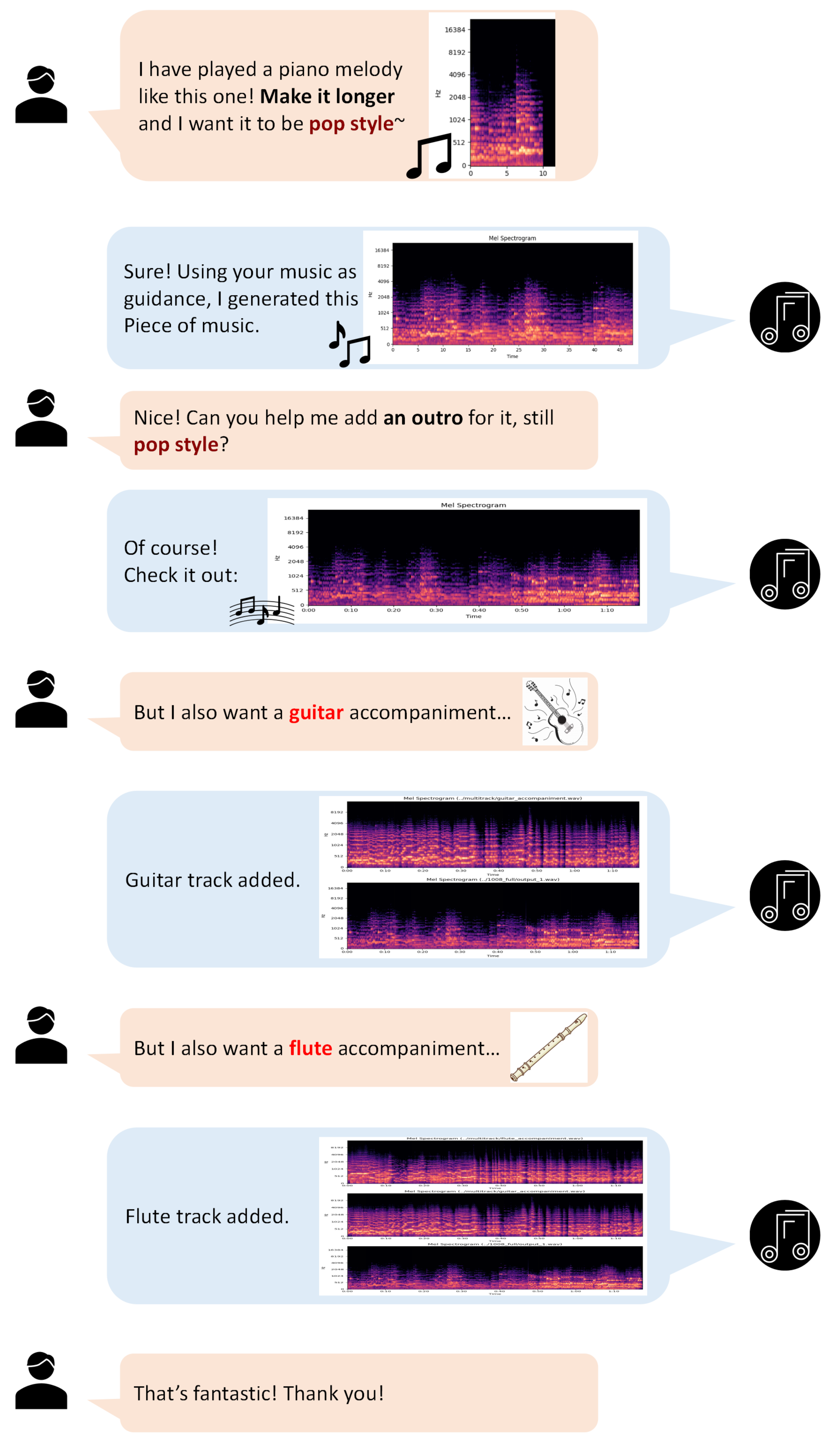

Figure 2: Music generation example.

Step	Harmony/Pitch	Rhythm	Timbre/Sound	Dynamics	Novelty	Continuation
224	49.37	17.79	99.21	99.10	67.48	0.5195
672	48.84	20.55	99.69	99.35	68.00	0.6913
1008	50.80	13.67	99.99	99.72	68.63	0.9463

Table 1: Average scores across different steps (highlighted: rhythm and continuation)

Also, we evaluate **DiffusionTunes** against stable-audio-open-1.0 [1] (without ControlNet) as baseline in multitrack generation task. The results are shown in Table 2.

Step	Harmony/Pitch	Rhythm	Dynamics	Novelty	Length
Base	47.60	19.63	97.60	67.69	48.0
672	48.84	20.55	99.35	68.00	48.0
Full	50.32	17.17	100.0	63.51	78.0
Multi	52.86	18.70	100.0	64.81	78.0

Table 2: Related scores for multitrack generation.

4 Conclusion & Discussion

We present **DiffusionTunes**, a controllable long-form music generation system that integrates melody and text conditioning through a three-stage architecture.

Our hierarchical designcombining diffusion generation, autoregressive semantic modeling, and track-wise orchestrationenables expressive and structurally consistent music outputs from short audio prompts. Future work may explore directions like accelerated generation, enabling vocal humming as input, etc.

References

[1] Zach Evans, Julian D. Parker, CJ Carr, Zack Zukowski, Josiah Taylor, and Jordi Pons. Stable audio open, 2024.
[2] Simon Rouard, Robin San Roman, Yossi Adi, and Axel Roebel. Musicgen-stem: Multi-stem music generation and edition through autoregressive modeling, 2025.
[3] Chong Zhang, Yukun Ma, Qian Chen, Wen Wang, Shengkui Zhao, Zexu Pan, Hao Wang, Chongjia Ni, Trung Hieu Nguyen, Kun Zhou, Yidi Jiang, Chaohong Tan, Zhifu Gao, Zhihao Du, and Bin Ma. Inspiremusic: Integrating super resolution and large language model for high-fidelity long-form music generation, 2025.